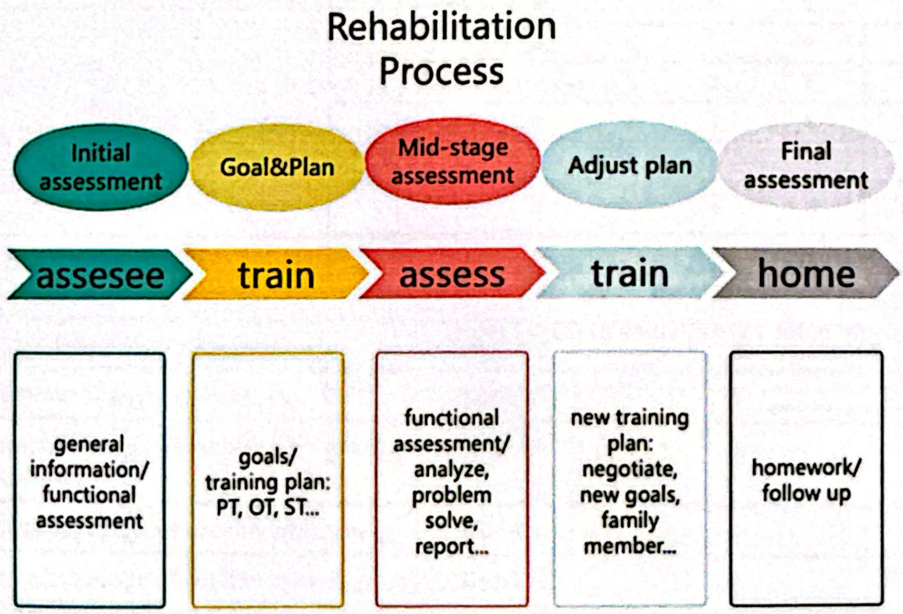
Functional Assessment Study Material

1. Introduction:

rehabilitation process:



case record information:

examples of Reliability and validity:

Reliability: Ensures consistent and stable results over time

Validity: Ensures the tool measures the intended construct accurately。

High reliability does not guarantee high validity, but validity requires reliability

2. Modified Barthel Index

改良 Barthel 指数评定量表（MBI）

项目	完全依赖	最大帮助	中等帮助	最小帮助	完全独立	每项评分
	1级	2级	3级	4级	5级	
(1) 进食	0	2	5	8	10	
(2) 个人卫生	0	1	3	4	5	
(3) 穿衣	0	2	5	8	10	
(4) 洗澡	0	1	3	4	5	
(5) 如厕	0	2	5	8	10	
(6) 大便控制	0	2	5	8	10	
(7) 小便控制	0	2	5	8	10	
(8) 床椅转移	0	3	8	12	15	
(9) 行走	0	3	8	12	15	
(9A) 轮椅操作*	0	1	3	4	5	
(10) 上下楼梯	0	2	5	8	10	

Modified Barthel Index for Activities of Daily Living

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Input

Chair/bed transfers	15
Ambulation	15
Ambulation/Wheelchair	0
Stair climbing	10
Toilet transfers	10
Bowel control	10
Bladder control	10
Bathing	5
Dressing	10
Personal Hygiene (Grooming)	5
Feeding	10

Result

Modified Barthel Index 100 points

Interpretation: 100 Independency

3. Muscle strength assessment: MRC/Lovett

Grade	Criteria
5	Normal strength
5-	Uncertain muscle weakness
4+	Ability to move through full range of motion and hold against strong pressure
4	Ability to move through full range of motion and hold against moderate pressure
4-	Ability to move through full range of motion and hold against slight pressure; or breaks abruptly with pressure
3	Ability to move through full range of motion against gravity
3-	Ability to move through partial range of motion against gravity
2	Ability to move through any range of motion only with gravity eliminated
1	A flicker of movement is seen or felt in the muscle
0	No contraction palpable

FUNCTIONAL LEVEL	LOVETT SCALE	GRADE	PERCENT OF NORMAL
No evidence of contractility	Zero (0)	0	0
Evidence of slight contractility	Trace (T)	1	10
Complete range of motion with gravity eliminated	Poor (P)	2	25
Complete range of motion with gravity	Fair (F)	3	50
Complete range of motion against gravity with some resistance	Good (G)	4	75
Complete range of motion against gravity with full resistance	Normal (N)	5	100

（上为 MRC；下为 LOVETT）

4. Muscle tone assessment: Modified Ashworth Scale

Modified Ashworth Scale Instructions

General Information (derived Bohannon and Smith, 1987):

- Place the patient in a supine position
- If testing a muscle that primarily flexes a joint, place the joint in a maximally flexed position and move to a position of maximal extension over one second (count "one thousand one")
- If testing a muscle that primarily extends a joint, place the joint in a maximally extended position and move to a position of maximal flexion over one second (count "one thousand one")
- Score based on the classification below

Scoring (taken from Bohannon and Smith, 1987):

0	No increase in muscle tone
1	Slight increase in muscle tone, manifested by a catch and release or by minimal resistance at the end of the range of motion when the affected part(s) is moved in flexion or extension
1+	Slight increase in muscle tone, manifested by a catch, followed by minimal resistance throughout the remainder (less than half) of the ROM
2	More marked increase in muscle tone through most of the ROM, but affected part(s) easily moved
3	Considerable increase in muscle tone, passive movement difficult
4	Affected part(s) rigid in flexion or extension

5. Human Morphology：

Cobb's angle and the treatment

Cobb angle	Treatment
cobb<25°	No treatment needed, follow up every 4-6 months for dynamic observation
25°<cobb<45°	Usually brace treatment
cobb>45°	Surgery recommended

measuring the length of limbs and stumps

measuring the circumference of limbs and stumps

Summary 明德致知 齐心济众

Physical assessment

1. Upper limb length: lateral end of the acromion → radial tuberosity or tip of the middle finger
2. Upper arm length: lateral end of acromion → lateral epicondyle of humerus
3. Forearm length: lateral epicondyle of humerus → radial tuberosity
4. Hand length: midpoint of the line between the radial styloid and ulnar styloid → tip of middle finger
5. Lower limb length: anterior superior iliac spine → medial ankle or greater trochanter → lateral ankle
6. Thigh length: greater trochanter of femur → lateral knee joint space
7. Lower leg length: lateral joint space of the knee → lateral ankle
8. Foot length: end of heel → end of second toe

Summary 明德致知 齐心济众

Measurement of amputation stump length (remember the measurement point)

1. Upper arm stump: anterior edge of the armpit → stump end
2. Forearm stump: ulnar humerus → end of stump
3. Thigh stump: sciatic tuberosity → end of stump
4. Lower leg stump: lateral knee joint gap → stump end

Summary

明德致知 齐心济众

Amputation stump circumference measurement

1. Upper arm stump circumference: anterior edge of the armpit → the end of the residual limb, every 2.5cm circumference measurement
2. Forearm stump circumference: ulnar humerus → residual limb end, every 2.5cm circumference measurement
3. Thigh stump circumference: sciatic tuberosity→end of the stump, measured every 5cm.
4. Calf stump circumference: lateral knee joint gap → the end of the stump, measured every 5cm circumference

Summary

明德致知 齐心济众

Amputation stump circumference measurement

Normal thigh circumference: measured at intervals of 6, 8, 10 and 12 cm from the upper edge of the patella to the mid-thigh.

The rest of the circumference: remember the measurement points, especially for chest, abdomen and hips.

6. Cardiopulmonary function assessment: Modified British Medical Research Council Dyspnea Scale, mMRC

MODIFIED MRC DYSPNEA SCALE[a]

PLEASE TICK IN THE BOX THAT APPLIES TO YOU | ONE BOX ONLY | Grades 0 - 4

mMRC Grade 0.	I only get breathless with strenuous exercise.	☐
mMRC Grade 1.	I get short of breath when hurrying on the level or walking up a slight hill.	☐
mMRC Grade 2.	I walk slower than people of the same age on the level because of breathlessness, or I have to stop for breath when walking on my own pace on the level.	☐
mMRC Grade 3.	I stop for breath after walking about 100 meters or after a few minutes on the level.	☐
mMRC Grade 4.	I am too breathless to leave the house or I am breathless when dressing or undressing.	☐

7. Normal range of joint motion: shoulder flexion/ shoulder extension.../

关节	活动度	关节	活动度
颈椎		腕	
屈曲	0°~45°	桡偏	0°~20°
伸展	0°~45°	掌屈	0°~80°
侧屈	0°~45°	背伸	0°~70°
旋转	0°~60°	尺偏	0°~30°
		旋转	0°~45°
胸腰椎		髋	
屈曲	0°~80°	屈曲	0°~120°
伸展	0°~30°	伸展	0°~30°/15°
侧屈	0°~40°	外展	0°~45°
旋转	0°~45°	内收	0°~35°
肩		内旋	0°~35°
屈曲	0°~170°	外旋	0°~45°
后伸	0°~60°	膝	
外展	0°~170°	屈曲	0°~135°
水平外展	0°~40°	踝	
水平内收	0°~130°	背屈	0°~15°
内旋	0°~70°	跖屈	0°~50°
外旋	0°~90°	内翻	0°~35°
肘和前臂		外翻	0°~20°
屈曲	0°~135°/150°		
旋后	0°~80°/90°		
旋前	0°~80°/90°		

Goniometer positioned when measuring joint motion:

Moving arm and fixed arm

测量关节运动时定位的测角仪： 动臂和固定臂

8. characteristic of Broca's Aphasia/Wernicke's Aphasia

[illegible]	自发语	流畅性	理解	复述	命名	损伤部位
Broca's失语	少，费力，电报式，失语法	受损	较好	受损	受损	后一下额叶
Wernicke's失语	赘语，错语，新语	正常	受损	受损	受损	后一上颞叶

Broca——运动型失语，以口语表达障碍为主

Wernicke——感觉性失语，有较多的错语或不易被人理解的新语

9. Types of sensation and how to measure sensation.

浅感觉——痛温触压觉

Pain， temperature，tactile sensation

深感觉——运动，位置，震动

Kinesthetic， joint position， vibration sensation

复合感觉——

Cutaneous Localization (皮肤定位觉)

Two-Point Discrimination (两点辨别觉)

Stereognosis (实体觉)

Graphesthesia (图形觉)

Barognosis(重量觉)

10. Grades of coordination assessment/ actions of coordination assessment

Grading

Level I	Normal completion
Level II	Mild impairment, able to complete the activity but with slight differences in speed and skill compared to normal
Level III	Moderate impairment, able to complete the activity but with slow, clumsy and obviously unstable movements
Level IV	Severe impairment, Only able to initiate the movement but unable to complete it
Level V	Unable to complete the activity

Sample Tests

Dysdiadochokinesia	Finger-to-nose Alternate nose-to-finger Pronation/supination Knee flexion/extension Walking, alter speed or direction
Dysmetria	Pointing and past pointing Drawing a circle or figure eight Heel on shin Placing feet on floor markers; sitting, standing
Dyssynergia	Finger-to-nose Finger-to-therapist's finger Alternate heel-to-knee Toe-to-examiner's finger

Tremor (intention)	Observation during functional activities (tremor will typically increase as target is approached or movement speed increased) Alternate nose-to-finger Finger-to-finger Finger-to-therapist's finger Toe-to-examiner's finger
Tremor (resting)	Observation of patient at rest; limb or jaw movements Observation during functional activities (tremor will diminish significantly or disappear with movement)

11. Brunnstrom Stage

Stage	Description
Stage 1	Lack of movement in the extremities.
Stage 2	Slight voluntary motor response in the extremities and the onset of spasticity.
Stage 3	Patients can control synkinesis autonomously, spasticity is severe.
Stage 4	Patients have control of detachment movements and spasticity begins to diminish.
Stage 5	The diminished role of co-movement and enhanced control of separate movement.
Stage 6	Normalization of movement and disappearance of spasticity.

12. assessment of spinal cord injury // the ASIA Impairment Scale

Assessment of injury level---key muscles

Right Side Score	Level	Representative Muscle	Left Side Score
5	C5	Biceps	5
5	C6	Extensor Carpi Radialis	5
	C7	Triceps	5
5	C8	Flexor Digitorum Profundus (Middle Finger)	5
5	|T1	Abductor Digiti Minimi (Little Finger)	5
5	L2	Iliopsoas	5
5	L3	Quadriceps	5
5	L4	Tibialis Anterior	5
5	L5	Extensor Hallucis Longus	5
5	S1	Gastrocnemius	5

ASIA Impairment Scale		
A	Complete	No motor , No sensory, No sacral sparing,
B	Incomplete	No motor, sensory only
C	Incomplete	50% of muscles LESS than grade 3 (cant not raise arms or legs off bed)
D	Incomplete	50% of muscles MORE than grade 3 (can raise arms or legs off bed)
E	Normal	Motor and sensory function are normal